

Exercise and worked solution

Interest rate products, bonds & swaps

Four parts to master the world of rates : rebuild a zero by bootstrap and derive a forward, measure price risk with duration and convexity, settle an FRA, and reason through comparative advantage then value a swap. Work through each part before reading the solution.

Part A · Bootstrap and implied forward

PROBLEM

(a) The semiannual zeros are 3% (6 months) and 4% (1 year). A 1.5-year bond, 4% semiannual coupon, face 100, quotes at 98.572. Find the 1.5-year zero. (b) With zeros of 3% (6 months) and 3.8% (1 year), compute the implied forward between 6 and 12 months.

SOLUTION

- (a) A single unknown R : $2/1.015 + 2/1.02^2 + 102/(1+R/2)^3 = 98.572$. The first two terms give 3.8928, so $102/(1+R/2)^3 = 94.679$, $(1+R/2)^3 = 1.07733$, $1+R/2 = 1.02516$, hence $R = \mathbf{5.03\%}$. The curve rises.
- (b) $V_1 = 1.015$, $V_2 = 1.019^2 = 1.038361$. The intermediate factor is $V_2/V_1 = 1.023016$, so $1 + F/2 = 1.023016$ and $F = \mathbf{4.6\%}$. The forward exceeds both zeros because the curve is upward-sloping.

Part B · Duration and convexity

PROBLEM

A bond has a Macaulay duration of 2.79 years and a yield of 5% compounded semiannually. (a) Compute the Modified duration. (b) Estimate the price change for a 25 bp yield rise. (c) Does convexity make the true fall larger or smaller?

SOLUTION

- (a) Modified = Macaulay/(1 + y/m) = $2.79/1.025 = \mathbf{2.72}$.
- (b) $\Delta B/B \approx -D \cdot \Delta y = -2.72 \times 0.0025 = \mathbf{-0.68\%}$: the price falls by about 0.68%.
- (c) Convexity is positive: the true fall is **smaller** than the linear estimate. The price/yield curve lies above its duration tangent, so duration alone overstates the loss on a rise and understates the gain on a fall. It is the holder's friend.

Part C · FRA settlement

PROBLEM

A firm locks 4% via an FRA on a 1M notional for a 3-month period. The realised floating rate is 4.4%. (a) Compute the settlement. (b) Why is it not simply the gross differential?

SOLUTION

- (a) Gross differential = $(0.044 - 0.04) \times 0.25 \times 1M = 1,000$. Since the FRA settles at the start of the period, discount at the floating rate: settlement = $1,000/(1 + 0.044 \times 0.25) = 1,000/1.011 = \mathbf{989.12}$.
- (b) Because interest is normally paid at the end of the period, but the FRA settles at the start: paying in advance is worth less, so one must discount. Forgetting this discounting is the exam's classic calculation trap.

Part D · Comparative advantage and swap valuation

PROBLEM

(a) A borrows at 3.5% fixed or $F + 0.1\%$; B at 4.9% fixed or $F + 0.9\%$. Total gain of a swap? Who borrows what? Is the gain always real? (b) An old swap receives $K = 4\%$ fixed, 2 years left, 100M, quarterly exchanges; $s = 2.96\%$, OIS 0.6%/quarter. Value it.

SOLUTION

1. (a) Fixed gap = 1.4%, floating gap = 0.8%, total gain = **0.6%**. A borrows fixed (largest advantage), B floating, then they swap. Caveat: the gain is partly illusory, since B's floating spread is locked for only a few months; if its credit degrades, its 'fixed' cost via the swap can exceed the direct fixed. Exception: currency swaps (genuine tax advantage).
2. (b) Overlay a new swap paying $s = 2.96\%$ (zero value): the floating legs cancel and the fixed annuity $\tau(K - s)L = 0.25 \times 1.04\% \times 100M = 260,000$ per quarter remains. Over 8 quarters, discounted at OIS 0.6%/quarter: $V = \Sigma 260,000/1.006^k =$ **2,024,945 USD**. Discounting is at OIS since the crisis (Libor before), as it reflects a near-risk-free rate consistent with collateral.

Key takeaways: the bootstrap rebuilds zeros one unknown at a time; the forward comes from the ratio of growth factors ($1 + F/2 = V_2/V_1$); Modified = Macaulay/($1 + y/m$) and $\Delta B/B \approx -D \cdot \Delta y$ for small parallel shifts, positive convexity being the holder's friend; the FRA settles at the start, so the differential is discounted; comparative advantage equals the gap of spreads but is real only if credit does not move; and a swap is valued by mirror swap, annuity $\tau(K - s)L$ discounted at OIS.